MC

$$\langle \mathcal{O}(q) \rangle = \int \mathcal{O}(q)\, p(q)\, d^{3N}q$$

$p(q)$: DISTRIBUTION NTV, NPT, GC

MD

$$\langle \mathcal{O}(t) \rangle = \frac{1}{\tau}\int_0^{\tau} \mathcal{O}(t)\, dt$$

TIME AVERAGE

MD MICROCANONICAL ENSEMBLE

HAMILTONIAN EQUATION

ISOLATED

NVE

$$\begin{cases} \dot{q}_i = \dfrac{\partial \mathcal{H}}{\partial q_i} \\ \dot{p}_i = \dfrac{\partial \mathcal{H}}{\partial p_i} \end{cases} \qquad i = 1 \dots N$$

$\frac{\partial \mathcal{H}}{\partial t} = 0$ $\mathcal{H}$ DOES NOT DEPEND EXPLICITLY ON TIME

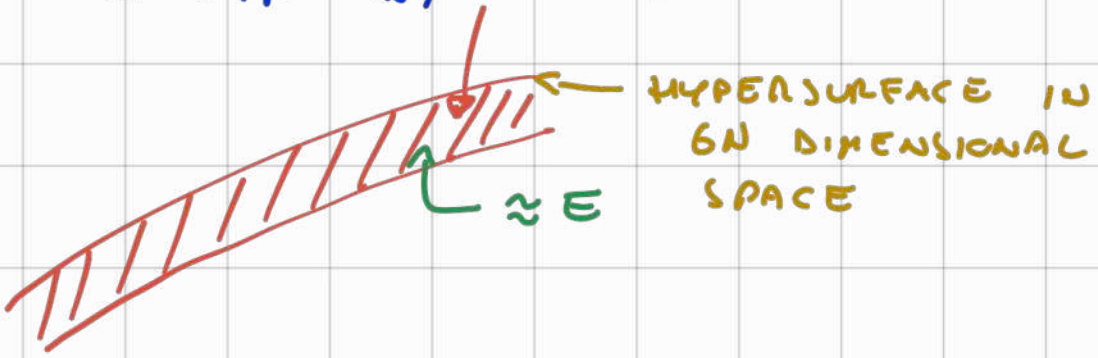
$$\mathcal{H}(q,p) = K + V = \sum_i^N \frac{\vec{p}_i^{\,2}}{2m} + V(\vec{q}_1 \dots \vec{q}_N)$$

$$\frac{d\mathcal{H}}{dt} = \sum_i \frac{\partial \mathcal{H}}{\partial \vec{q}_i}\dot{\vec{q}}_i + \frac{\partial \mathcal{H}}{\partial \vec{p}_i}\dot{\vec{p}}_i =$$

$$= \sum_i \frac{\partial \mathcal{H}}{\partial \vec{q}_i}\frac{\partial \mathcal{H}}{\partial \vec{p}_i} - \frac{\partial \mathcal{H}}{\partial \vec{p}_i}\frac{\partial \mathcal{H}}{\partial \vec{q}_i} = 0$$

ENERGY IS CONSERVED

$$\mathcal{H}(\vec{q}_1 \dots \vec{q}_N, \vec{p}_1 \dots \vec{p}_N) \approx E$$



$$\vec{P} = \sum_i \vec{p}_i$$

TOTAL MOMENTUM IS ALSO CONSERVED

$$\vec{L} = \sum_i \vec{r}_i \times \vec{p}_i$$

TOTAL ANGULAR MOMENTUM IS NOT CONSERVED IF WE RESTRICT TO SIMULATION BOX CENTERED AT ORIGIN

LAGRANGIAN FORMULATION

$$\mathcal{L} = K - V = \mathcal{L}(\vec{q}_1 \dots \vec{q}_N, \dot{\vec{q}}_1 \dots \dot{\vec{q}}_N)$$

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{\vec{q}}_i}\right) - \frac{\partial \mathcal{L}}{\partial \vec{q}_i} = 0 \Rightarrow \vec{F}_i = m_i \vec{a}_i$$

$$\ddot{\vec{q}}_i = \frac{\vec{F}_i}{m_i} \qquad i = 1 \dots N$$

LIOUVILLE'S THEOREM

HYDRODYNAMICS :

$\vec{v}(x)$ $\quad \vec{j} = \rho \vec{v}$

$$\nabla \cdot \vec{j} + \frac{\partial \rho}{\partial t} = 0$$ CONTINUITY EQUATION

NOTE THAT

$$\nabla \cdot \vec{j} + \frac{\partial \rho}{\partial t} = \frac{\partial \rho}{\partial \vec{x}} \cdot \vec{v} + \rho \nabla \cdot \vec{v} + \frac{\partial \rho}{\partial t} =$$

IF WE DEFINE $\quad \frac{D\rho}{Dt} \equiv \frac{\partial \rho}{\partial \vec{x}} \cdot \vec{v} + \frac{\partial \rho}{\partial t}$ MATERIAL DERIVATIVE

$$\nabla \cdot \vec{j} + \frac{\partial \rho}{\partial t} = \frac{D\rho}{Dt} + \rho \nabla \cdot \vec{v} = 0$$

HENCE IF THE FLUID IS INCOMPRESSIBLE

$$\frac{D\rho}{Dt} = 0$$ WHICH IMPLIES

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \vec{v} = \rho \nabla \cdot \vec{v} = 0 \Rightarrow \nabla \cdot \vec{v} = 0$$

AND ALSO THE OTHER WAY AROUND, I.E.

IF $\nabla \cdot \vec{v} = 0 \Rightarrow \frac{D\rho}{Dt} = 0$ I.E. THE FLUID IS INCOMPRESSIBLE

MAKING AN ANALOGY WITH HYDRODYNAMICS WE HAVE

$$\vec{v} = (\dot{\vec{q}}_1 \dots \dot{\vec{q}}_N, \dot{\vec{p}}_1 \dots \dot{\vec{p}}_N)$$

HENCE IF $\nabla \cdot \vec{v} = \sum_i^N \frac{\partial \dot{\vec{q}}_i}{\partial \vec{q}_i} + \sum_i^N \frac{\partial \dot{\vec{p}}_i}{\partial \vec{p}_i}$

BUT FOR AN HAMILTONIAN SYSTEM

$$\nabla \cdot \vec{v} = \sum_{i=1}^N \frac{\partial \dot{\vec{q}}_i}{\partial \vec{q}_i} - \frac{\partial \dot{\vec{p}}_i}{\partial \vec{p}_i} = 0$$

$K = \nabla \cdot \vec{v}$ IS CALLED COMPRESSIBILITY

WHICH IS THE LIOUVILLE'S THEOREM.

THIS IMPLIES THAT VOLUMES IN PHASE SPACE ARE CONSERVED WHEN THEY EVOLVE ACCORDING TO HAMILTONIAN EQUATIONS OF MOTION (SINCE ρ IS CONSTANT)

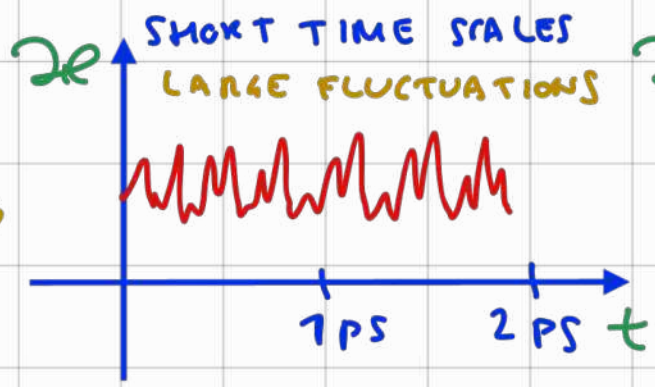
NUMERICAL INTEGRATION

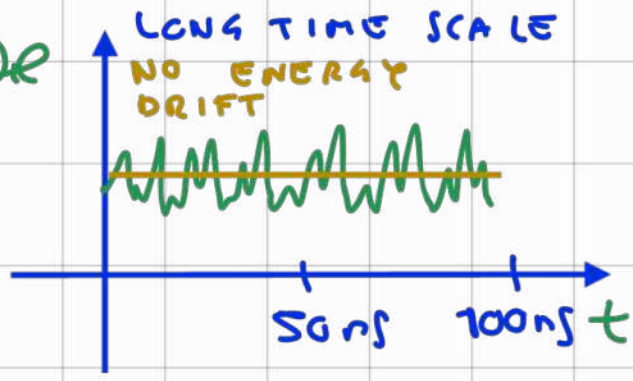
WE WANT TO USE FINITE DIFFERENCE METHODS

$$\vec{v}_i(t + \Delta t) = \vec{v}_i(t) + \vec{a}_i \Delta t + O(\Delta t^2)$$

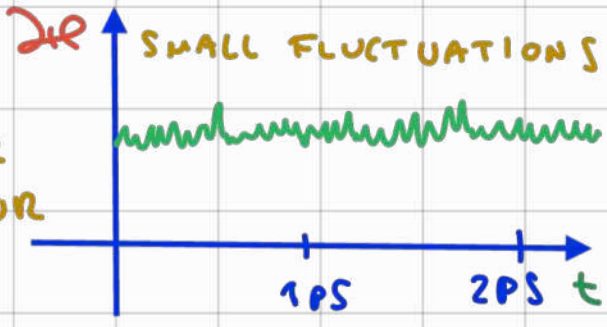
THERE ARE TWO GENERAL ASPECTS TO TAKE INTO ACCOUNT FOR AN ALGORITHM

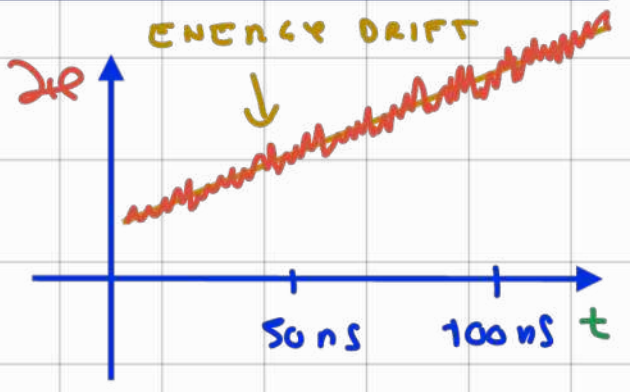
- VERLET
- VELOCITY VERLET
- LEAP-FROG





- GEAR PREDICTOR CORRECTOR





ENERGY DRIFT IS MORE IMPORTANT FOR HAVING ACCURATE TIME AVERAGES (IF ENERGY CHANGES IS LIKE HEATING UP OR COOLING DOWN THE SYSTEM)

VERLET, VELOCITY VERLET AND LEAP-FROG GUARANTEE A NEGLIGIBLE ENERGY DRIFT

VERLET ALGORITHM

① $\vec{r}_i(t+\Delta t) = \vec{r}_i(t) + \Delta t\, \vec{v}_i(t) + \frac{\Delta t^2}{2}\vec{a}_i(t) + O(\Delta t^3)$

② $\vec{r}_i(t-\Delta t) = \vec{r}_i(t) - \Delta t\, \vec{v}_i(t) + \frac{\Delta t^2}{2}\vec{a}_i(t) + O(\Delta t^3)$

IF WE SUM ① AND ② WE HAVE

$$\vec{r}_i(t+\Delta t) + \vec{r}_i(t-\Delta t) = 2\vec{r}_i(t) + \Delta t^2 \vec{a}_i(t) + \\ + \mathcal{O}(\Delta t^4)$$

$$\vec{r}_i(t+\Delta t) = 2\vec{r}_i(t) - \vec{r}_i(t-\Delta t) + \Delta t^2 \vec{a}_i(t) + \mathcal{O}(\Delta t^4)$$

$$\vec{a}_i = \frac{\vec{F}_i}{m_i} \leftarrow \text{TOTAL FORCE ACTING ON PARTICLE } i$$

$$\vec{F}_i = \vec{F}_i(\vec{r}_1 \dots \vec{r}_N)$$

WITH

$$\vec{F}_e = -\frac{\partial}{\partial \vec{r}_e}\left(\sum_{i<j} v_2(|\vec{r}_i - \vec{r}_j|)\right)$$

INSTEAD IF WE SUBTRACT ① AND ② :

$$\vec{v}_i(t) = \frac{\vec{r}_i(t+\Delta t) - \vec{r}_i(t-\Delta t)}{2\Delta t} + \mathcal{O}(\Delta t^2)$$

THUS HAVING THE VELOCITY AT TIME t AND THE VERLET ALGORITHM READS :

$$\vec{r}_i(t+\Delta t) = 2\vec{r}_i(t) - \vec{r}_i(t-\Delta t) + \Delta t^2 \vec{a}_i(t) \qquad (V1)$$

$$\vec{v}_i(t) = \frac{\vec{r}_i(t+\Delta t) - \vec{r}_i(t-\Delta t)}{2\Delta t} \qquad (V2)$$

LEAP-FROG

$$\vec{r}_i(t+\Delta t) = \vec{r}_i(t) + \Delta t\, \vec{v}_i(t) + \frac{\Delta t^2}{2}\vec{a}_i(t) + \mathcal{O}(\Delta t^3)$$

$$\Delta t\, \vec{v}_i\left(t+\frac{\Delta t}{2}\right) = \Delta t\, \vec{v}_i(t) + \frac{\Delta t^2}{2}\vec{a}_i(t) + \mathcal{O}(\Delta t^3)$$

HENCE

FIRST EQUATION OF LEAP-FROG

$$\vec{r}_i(t+\Delta t) = \vec{r}_i(t) + \Delta t\, \vec{v}_i\left(t+\frac{\Delta t}{2}\right) + \mathcal{O}(\Delta t^3)$$

$$① \quad \vec{v}_i\left(t+\frac{\Delta t}{2}\right) = \vec{v}_i(t) + \frac{\Delta t}{2}\vec{a}_i(t) + \mathcal{O}(\Delta t^2)$$

$$② \quad \vec{v}_i\left(t-\frac{\Delta t}{2}\right) = \vec{v}_i(t) - \frac{\Delta t}{2}\vec{a}_i(t) + \mathcal{O}(\Delta t^2)$$

① − ②

$$\vec{v}_i\left(t+\frac{\Delta t}{2}\right) - \vec{v}_i\left(t-\frac{\Delta t}{2}\right) = \Delta t\, \vec{a}_i(t) + \mathcal{O}(\Delta t^3)$$

$$\vec{v}_i\left(t+\frac{\Delta t}{2}\right) = \vec{v}_i\left(t-\frac{\Delta t}{2}\right) + \Delta t\, \vec{a}_i(t) + \mathcal{O}(\Delta t^3)$$

↑ SECOND EQUATION OF LEAP-FROG

SO THAT THE LEAP-FROG ALGORITHM IS:

$$\vec{r}_i(t+\Delta t) = \vec{r}_i(t) + \Delta t\, \vec{v}_i\left(t+\frac{\Delta t}{2}\right) \qquad \text{(LF1)}$$

$$\vec{v}_i\left(t+\frac{\Delta t}{2}\right) = \vec{v}_i\left(t-\frac{\Delta t}{2}\right) + \Delta t\, \vec{a}_i(t) \qquad \text{(LF2)}$$

VELOCITY VERLET

① $$\vec{r}_i(t+\Delta t) = \vec{r}_i(t) + \Delta t\, \vec{v}_i(t) + \frac{\Delta t^2}{2}\vec{a}_i(t) + \frac{\Delta t^3}{6}\dot{\vec{a}}_i(t) + \mathcal{O}(\Delta t^4)$$

② $$\vec{r}_i(t) = \vec{r}_i((t+\Delta t)-\Delta t) =$$

$$\vec{r}_i(t+\Delta t) - \Delta t\, \vec{v}_i(t+\Delta t) + \frac{\Delta t^2}{2}\vec{a}_i(t-\Delta t) - \frac{\Delta t^3}{6}\dot{\vec{a}}_i(t-\Delta t) + \mathcal{O}(\Delta t^4) =$$

$$\vec{r}_i(t+\Delta t) - \Delta t\, \vec{v}_i(t+\Delta t) + \frac{\Delta t^2}{2}\vec{a}_i(t-\Delta t) - \frac{\Delta t^3}{6}\vec{a}_i(t) + \mathcal{O}(\Delta t^4)$$

① + ②:

$$\vec{r}_i(t+\Delta t) + \vec{r}_i(t) =$$

$$\vec{r}_i(t+\Delta t) + \vec{r}_i(t) + \Delta t\, \vec{v}_i(t) - \Delta t\, \vec{v}_i(t+\Delta t) + \frac{\Delta t^2}{2}\left(\vec{a}_i(t) + \vec{a}_i(t+\Delta t)\right) + \frac{\Delta t^3}{6}\dot{\vec{a}}_i(t) - \frac{\Delta t^3}{6}\dot{\vec{a}}_i(t) + \mathcal{O}(\Delta t^4)$$

SECOND EQUATION

$$\vec{v}_i(t+\Delta t) = \vec{v}_i(t) + \frac{\Delta t}{2}\left(\vec{a}_i(t) + \vec{a}_i(t+\Delta t)\right) + \mathcal{O}(\Delta t^3)$$

FROM (1) BY CONSIDERING TERMS UP TO $\mathcal{O}(\Delta t^2)$:

$$\vec{r}_i(t+\Delta t) = \vec{r}_i(t) + \Delta t\,\vec{v}_i(t) + \frac{\Delta t^2}{2}\vec{a}_i(t) + \mathcal{O}(\Delta t^3)$$

FIRST EQUATION OF VELOCITY VERLET

HENCE VELOCITY VERLET ALGORITHM IS:

$$\vec{r}_i(t+\Delta t) = \vec{r}_i(t) + \Delta t\,\vec{v}_i(t) + \frac{\Delta t^2}{2}\vec{a}_i(t) \qquad (VV1)$$

$$\vec{v}_i(t+\Delta t) = \vec{v}_i(t) + \frac{\Delta t}{2}\left(\vec{a}_i(t) + \vec{a}_i(t+\Delta t)\right) \qquad (VV2)$$

THESE THREE ALGORITHMS ARE ALGEBRICALLY EQUIVALENT

INDEED THEY ARE ALL DERIVED FROM A TAYLOR EXPANSION OF $\vec{r}_i(t+\Delta t)$ UP TO SECOND ORDER

VERLET IS EQUIVALENT TO VELOCITY VERLET

FROM (V1):

$$\vec{r}_i(t+\Delta t) = 2\vec{r}_i(t) - \vec{r}_i(t-\Delta t) + \Delta t^2\,\vec{a}_i(t)$$

$$\vec{r}_i(t+\Delta t) = 2\vec{r}_i(t) - \vec{r}_i(t+\Delta t) + \vec{r}_i(t+\Delta t) - \vec{r}_i(t-\Delta t) + \Delta t^2\,\vec{a}_i(t)$$

BY USING (V2):

$$2\vec{r}_i(t+\Delta t) = 2\vec{r}_i(t) + 2\Delta t\,\vec{v}_i(t) + \Delta t^2\,\vec{a}_i(t)$$

$$\vec{r}_i(t+\Delta t) = \vec{r}_i(t) + \Delta t\,\vec{v}_i(t) + \frac{\Delta t^2}{2}\vec{a}_i(t)$$

WHICH IS (VV1)

CONCERNING (VV2) WE NEED TO PROVE THAT

$$\vec{v}_i(t+\Delta t) - \vec{v}_i(t) = \frac{\Delta t}{2}\left(\vec{a}_i(t) + \vec{a}_i(t+\Delta t)\right)$$

AND THIS CAN BE DONE BY USING EQ.(V2), I.E.

$$\vec{v}_i(t+\Delta t) - \vec{v}_i(t) =$$

$$\frac{\vec{r}_i(t+2\Delta t) - \vec{r}_i(t)}{2\Delta t} - \frac{\vec{r}_i(t+\Delta t) - \vec{r}_i(t-\Delta t)}{2\Delta t}$$

AND NOW WE CAN USE EQ.(V1) FOR BOTH $\vec{r}_i(t+2\Delta t)$ AND $\vec{r}_i(t-\Delta t)$ TO HAVE

$$2\vec{r}_i(t+\Delta t) - \vec{r}_i(t) - \vec{r}_i(t) + \Delta t^2 \vec{a}_i(t+\Delta t) +$$

$$- \vec{r}_i(t+\Delta t) + 2\vec{r}_i(t) - \vec{r}_i(t+\Delta t) + \Delta t^2 \vec{a}_i(t) =$$

$$= \Delta t^2 \left(\vec{a}_i(t) + \vec{a}_i(t+\Delta t)\right)$$

SO THAT

(*) FROM (V1): $\vec{r}_i(t-\Delta t) = 2\vec{r}_i(t) - \vec{r}_i(t+\Delta t) + \Delta t^2 \vec{a}_i(t)$

$$\vec{v}_i(t+\Delta t) - \vec{v}_i(t) = \frac{\Delta t}{2}\left(\vec{a}_i(t) + \vec{a}_i(t+\Delta t)\right)$$

WHICH IS (VV2)

LEAP-FROG IS EQUIVALENT TO VELOCITY-VERL.

WE WANT TO SHOW FIRST THAT

$$\vec{v}_i(t+\Delta t) - \vec{v}_i(t) = \frac{\Delta t}{2}\left(\vec{a}_i(t) + a_i(t+\Delta t)\right)$$

BUT FROM EQ.(LF2) BY ADVANCING BY HALF STEP

$$\vec{v}_i(t+\Delta t) = \vec{v}_i\left(t+\frac{\Delta t}{2}\right) + \frac{\Delta t}{2}\vec{a}_i(t+\Delta t) \quad (1)$$

SIMILARLY BY GOING BACK HALF TIME FROM $t+\frac{\Delta t}{2}$

$$\vec{v}_i(t) = \vec{v}_i\left(t+\frac{\Delta t}{2}\right) - \frac{\Delta t}{2}\vec{a}_i(t) \text{ I.E.}$$

$$\vec{v}_i(t+\Delta t/2) = \vec{v}_i(t) + \frac{\Delta t}{2}\vec{a}_i(t) \qquad (2)$$

HENCE IF WE SUM (1) AND (2):

$$\vec{v}_i(t+\Delta t) = \vec{v}_i(t) + \frac{\Delta t}{2}\left(\vec{a}_i(t) + \vec{a}_i(t+\Delta t)\right)$$

WHICH IS EQ. (VV2)

CONCERNING EQ.(VV1) WE NOTE THAT EQ.(LF1) IS:

$$\vec{r}_i(t+\Delta t) = \vec{r}_i(t) + \Delta t\, \vec{v}_i\left(t + \frac{\Delta t}{2}\right)$$

BUT FROM EQ.(2):

$$\vec{r}_i(t+\Delta t) = \vec{r}_i(t) + \Delta t\, \vec{v}_i(t) + \frac{\Delta t^2}{2}\vec{a}_i(t)$$

WHICH IS EQ. (VV1)

ENERGY CONSERVATION

$$\mathcal{H}(t) = \sum_i \frac{1}{2} m \vec{v}_i^2(t) + V(\vec{r}_1(t) .. \vec{r}_N(t))$$

WHERE $\{\vec{r}_i(t), \vec{v}_i(t)\}_i$ IS AN EXACT SOLUTION

IF FROM A FD ALGORITHM WE OBTAIN $\vec{r}_i^{FD}(\Delta t)$ AND $\vec{v}_i^{FD}(\Delta t)$ AS FOLLOWS:

$$\vec{r}_i^{FD}(\Delta t) = \vec{r}_i(\Delta t) + \mathcal{O}(\Delta t^3) \qquad (2)$$

(EXACT SOLUTION)

$$\vec{v}_i^{FD}(\Delta t) = \vec{v}_i(\Delta t) + \mathcal{O}(\Delta t^3) \qquad (3)$$

IF $\mathcal{H}^{FD}(\Delta t) \equiv \mathcal{H}(\vec{r}_1^{FD}(\Delta t) .. \vec{r}_N^{FD}(\Delta t), \vec{v}_1^{FD}(\Delta t) .. \vec{v}_N^{FD}(\Delta t))$

BY USING EQS. (2) AND (3) ONE HAS

$$\mathcal{H}^{FD}(\Delta t) = \mathcal{H}(\vec{r}_1(\Delta t) + \mathcal{O}(\Delta t^3) \ldots \vec{v}_N(\Delta t) + \mathcal{O}(\Delta t^3)) =$$

AND AFTER n STEPS:

$$\mathcal{H}^{FD}(n\Delta t) = \mathcal{H}(\vec{r}_1(n\Delta t) + n\mathcal{O}(\Delta t^3) \ldots \vec{v}_N(n\Delta t) + n\mathcal{O}(\Delta t^3)) \approx$$

$\simeq \mathcal{H}(\vec{r}_1(n\Delta t)\ldots\vec{p}_N(n\Delta t)) + \underbrace{n\,\mathcal{O}(\Delta t^3)}_{\Delta\mathcal{H}}$

NOTE THAT THIS IS CORRECT ONLY IF $n\,\mathcal{O}(\Delta t^3) \ll 1$

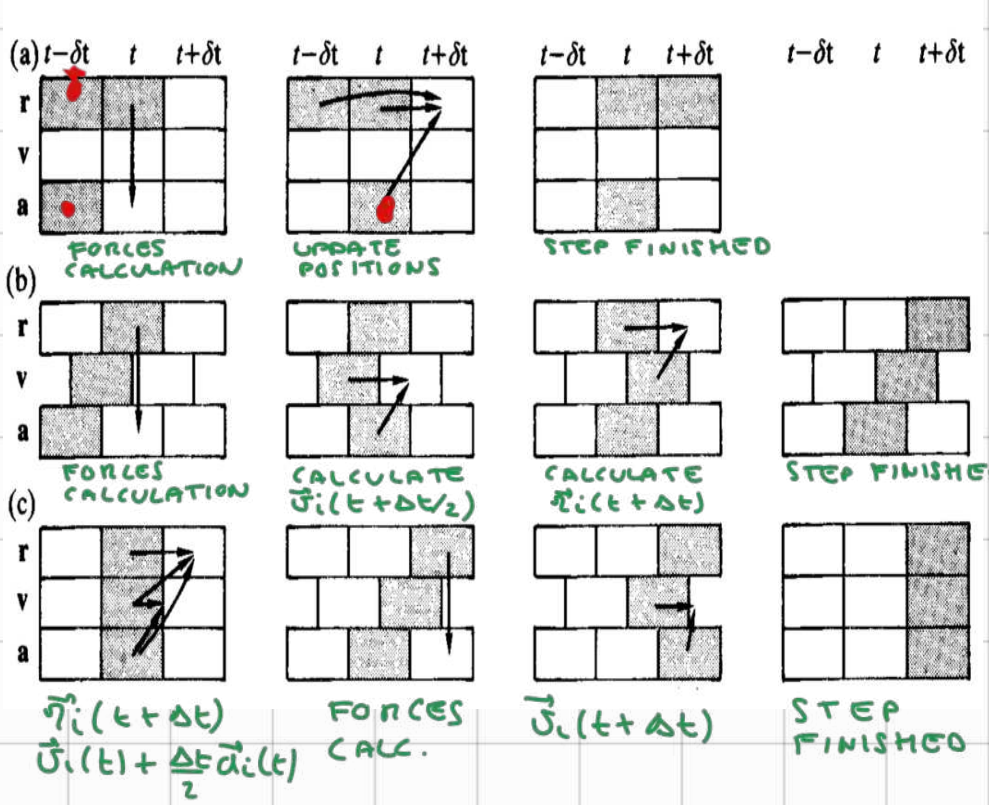
I.E. FOR SHORT TIMES (SEE BELOW)

THIS IMPLIES THAT NOT NECESSARILY FLUCTUATIONS OF ENERGY INCREASES WITH TIME.

INDEED WE WILL SHOW BY THE SHADOW HAMILTONIAN THAT THEY ARE EXACTLY $\mathcal{O}(\Delta t^2)$

HENCE

$$\Delta\mathcal{H} = n\,\mathcal{O}(\Delta t^3) = n\Delta t\,\mathcal{O}(\Delta t^2) = t\,\underbrace{\mathcal{O}(\Delta t^2)}_{\text{QUADRATIC IN } \Delta t}$$



VERLET

$\vec{r}_i(t+\Delta t) = 2\vec{r}_i(t) - \vec{r}_i(t-\Delta t) + \Delta t^2\vec{a}_i(t)$

LEAP-FROG

$\vec{r}_i(t+\Delta t) = \vec{r}_i(t) + \Delta t\,\vec{p}_i(t+\frac{\Delta t}{2})$

$\vec{p}_i(t+\frac{\Delta t}{2}) = \vec{p}_i(t-\frac{\Delta t}{2}) + \Delta t\,\vec{a}_i(t)$

VELOCITY-VERLET

$\vec{r}_i(t+\Delta t) = \vec{r}_i(t) + \Delta t\,\vec{p}_i(t) + \frac{\Delta t^2}{2}\vec{a}_i(t)$

$\vec{p}_i(t+\Delta t) = \vec{p}_i(t) + \frac{\Delta t}{2}(\vec{a}_i(t) + \vec{a}_i(t+\Delta t))$

SIMPLECTIC CONDITION

$$\dot{\vec{q}}_i = \frac{\partial \mathcal{H}}{\partial \vec{p}_i}$$

$$\dot{\vec{p}}_i = -\frac{\partial \mathcal{H}}{\partial \vec{q}_i}$$

$$\vec{Q} \equiv (\vec{q}_1 \dots \vec{q}_N) \qquad \vec{P} \equiv (\vec{p}_1 \dots \vec{p}_N)$$

$$X \equiv \begin{pmatrix} \vec{Q} \\ \vec{P} \end{pmatrix}$$ COLUMN VECTOR

THEN WE CAN REWRITE HAMILTON EQS. AS FOLLOWS:

$$\dot{X} = J^t \left(\frac{\partial \mathcal{H}}{\partial X}\right)$$ ← TRANSPOSE

WITH

$$J = \begin{pmatrix} 0 & \mathbb{1} \\ -\mathbb{1} & 0 \end{pmatrix}$$

$\mathbb{1}$ = 3N x 3N IDENTITY

J = 6N x 6N MATRIX

$\frac{\partial}{\partial X}$ IS A ROW VECTOR

COORDINATE TRANSFORMATION

$$Y = Y(X)$$

IF THIS TRANSFORMATION IS CANONICAL FORM OF HAMILTON EQS. MUST BE UNCHANGED, I.E.

$$\dot{\psi} = J^{t}\left(\frac{\partial \mathcal{H}}{\partial \psi}\right)$$

BUT

$$\dot{\psi} = \frac{d}{dt}(\psi(x)) = \frac{\partial \psi}{\partial x}\,\dot{x}$$

$$\frac{\partial \psi}{\partial x} = \begin{pmatrix} \frac{\partial \psi_1}{\partial x_1} & \cdots & \frac{\partial \psi_1}{\partial x_N} \\ \vdots & & \\ \frac{\partial \psi_N}{\partial x_1} & \cdots & \frac{\partial \psi_N}{\partial x_N} \end{pmatrix}$$

JACOBIAN OF TRANSFORMATION

$$\dot{x} = J^{t}\left(\frac{\partial \mathcal{H}}{\partial x}\right)$$

HENCE

$$\dot{\psi} = \frac{\partial \psi}{\partial x}\, J^{t}\left(\frac{\partial \mathcal{H}}{\partial x}\right)$$

$$\frac{\partial \mathcal{H}}{\partial x} = \frac{\partial \mathcal{H}}{\partial \psi}\,\frac{\partial \psi}{\partial x}$$ (CHAIN RULE)

THEREFORE

$$\dot{\eta} = \frac{\partial \varphi}{\partial x} J^t \left(\frac{\partial \mathcal{H}}{\partial \varphi} \frac{\partial \varphi}{\partial x} \right) =$$

$$= \frac{\partial \varphi}{\partial x} J^t \left(\frac{\partial \varphi}{\partial x} \right) {}^t \left(\frac{\partial \mathcal{H}}{\partial \varphi} \right)$$

IF TRANSFORMATION IS CANONICAL:

$$\frac{\partial \varphi}{\partial x} J^t \left(\frac{\partial \varphi}{\partial x} \right) = J$$

IF $\frac{\partial \varphi}{\partial x} = M$

$$\boxed{M J {}^t M = J}$$ SIMPLECTIC CONDITION

IF THIS CONDITION HOLDS, INDEED

$$\dot{\eta} = J^t \left(\frac{\partial \mathcal{H}}{\partial \varphi} \right)$$

LET'S CONSIDER THE MAPPING (TRANSFORMATION) INDUCED BY THE SOLUTION OF HAMILTON'S EQUATIONS, I.E.

$$\underset{\parallel}{X(t)} = \phi \left(t, \underset{\parallel}{X_0} \right)$$
$$\eta \qquad\qquad X$$

$$\eta = \varphi(x)$$

THIS MAPPING IS CANONICAL

PROOF (POINCARE' THEOREM)

$X(t) = \phi(t, X_0)$ TIME EVOLUTION ACCORDING TO HAMILTON EQS.

$X_0 \rightarrow X(t)$ MAPPING

$M(t) = \dfrac{\partial \phi(t, X_0)}{\partial X_0}$ JACOBIAN OF TRANSFORMATION

SYMPLECTIC CONDITION READS

$$M(t)\, J\, {}^{t}M(t) = J \qquad (C1)$$

AT $t=0$ THIS CONDITION IS FULFILLED, SINCE

$$M(0) = \left.\frac{\partial \phi(t, X_0)}{\partial X_0}\right|_{t=0} = \mathbb{1}$$

SO THAT (C1) REDUCES TO

$$J = J$$

HENCE, IF WE PROVE THAT :

$$\frac{d}{dt}\left(M(t)\, J\, {}^{t}M(t) \right) = 0 \qquad \forall t \qquad (P1)$$

THIS ENSURES THAT CONDITION (C1) IS FULFILLED $\forall t$

FIRST WE PROVE THAT CONDITION (C1) IS EQUIVALENT TO :

$${}^{t}M\, J\, M = J \qquad (C2)$$

INDEED

$$J = \begin{pmatrix} 0 & \mathbb{1} \\ -\mathbb{1} & 0 \end{pmatrix} \Rightarrow J^2 = \begin{pmatrix} -\mathbb{1} & 0 \\ 0 & -\mathbb{1} \end{pmatrix} = -\mathbb{1}$$

AND

$$^tJ\,J = \begin{pmatrix} \mathbb{1} & 0 \\ 0 & \mathbb{1} \end{pmatrix} = \mathbb{1}$$

$$M J\, {}^tM = J$$

$$M J\, \underbrace{{}^tM ({}^tM)^{-1}}_{\mathbb{1}} = J({}^tM)^{-1}$$

$$MJ = J({}^tM)^{-1} \qquad (C3)$$

MOREOVER CONSIDER

$$\boxed{J(MJ)J} = JMJ^2 = \boxed{-JM} \qquad (CA)$$

BY USING (C3) :

$$\boxed{J(MJ)J} = J(J({}^tM)^{-1})J =$$

$$\underbrace{JJ}_{-\mathbb{1}}({}^tM)^{-1}J = \boxed{-({}^tM)^{-1}J} \qquad (C5)$$

FROM (CA) AND (C5):

$$-JM = -({}^tM)^{-1}J$$

I.E.

$${}^tM J M = {}^tM \cancel{({}^tM)^{-1}}\,{}^{\mathbb{1}}J$$

$$\Downarrow$$

$$\boxed{{}^tM J M = J}$$

HENCE EQ. (P1) IS EQUIVALENT TO:

$$\frac{d}{dt}\left({}^t M(t)\, J\, M(t)\right) = 0$$

I.E.

$${}^t\dot{M}\, J\, M + {}^t M J \dot{M} = 0 \qquad (C6)$$

BUT

$$\dot{M} = \frac{d}{dt}[M(t)] = \frac{d}{dt}\left[\frac{\partial \phi(t,x_0)}{\partial x_0}\right] = \frac{\partial}{\partial x_0}\left[\dot{\phi}(t,x_0)\right]$$

SINCE $\phi(t,x_0)$ IS A SOLUTION OF HAMILTON'S EQUATIONS:

$$\dot{\phi}(t,x_0) = J\, {}^t\left(\frac{\partial \mathcal{H}}{\partial x}\right)_{x=\phi(t,x_0)}$$

REMEMBER THAT $x = (\vec{q}_1 \dots \vec{q}_N, \vec{p}_1 \dots \vec{p}_N)$

$$\frac{\partial}{\partial x_0}\left[\dot{\phi}(t,x_0)\right] = \frac{\partial}{\partial x_0}\left[J\, {}^t\left(\frac{\partial \mathcal{H}}{\partial x}\right)_{x=\phi(t,x_0)}\right] =$$

$$= J \left.\frac{\partial^2 \mathcal{H}}{\partial x\, \partial x}\right|_{x=\phi(t,x_0)} \frac{\partial \phi}{\partial x_0} = J\, H_{xx}\, M$$

$\frac{\partial \phi}{\partial x_0} = M$

${}^t\left(\frac{\partial \mathcal{H}}{\partial x}\right)$ IS A COLUMN VECTOR, HENCE $\frac{\partial}{\partial x}\left[{}^t\left(\frac{\partial \mathcal{H}}{\partial x}\right)\right]$ IS A MATRIX

WHERE

$$H_{xx} = \left.\frac{\partial^2 \mathcal{H}}{\partial x\, \partial x}\right|_{x=\phi(t,x_0)}$$

THEREFORE

$$\dot{M} = J\, H_{xx}\, M \qquad \text{AND} \qquad (C7)$$

$${}^t\dot{M} = {}^t M\, H_{xx}\, {}^t J \qquad (H_{xx} = {}^t H_{xx}) \quad (C8)$$

CONSIDER EQ. ((C), I.E.

$$ {}^t\dot{M}\, J\, M + {}^t M\, J\, \dot{M} = 0 $$

WHICH BY USING EQS. ((7) AND ((8) BECOMES:

$$ {}^t M\, H_{xx}\, \underbrace{{}^t J\, J}_{-1}\, M + {}^t M\, \underbrace{J\, J}_{-1}\, H_{xx}\, M = $$

$$ = {}^t M\, H_{xx}\, M - {}^t M\, H_{xx}\, M = 0 $$

THUS PROVING THE POINCARÈ THEOREM

MAPPING INDUCED BY SOLUTION OF HAMILTON'S EQUATION'S IS CANONICAL

LIOUVILLE'S THEOREM (AGAIN)

IF $\phi(t, X_0)$ IS A SOLUTION OF HAMILTON'S EQUATIONS AND

AND $M = \dfrac{\partial \phi(t, X_0)}{\partial X_0}$ FROM POINCARÈ'S THEOREM IT FOLLOWS THAT:

$$ M\, J\, {}^t M = J $$

BUT

$$ \det(M\, J\, {}^t M) = \det(J) $$

$$ \det(M)\, \det(J)\, \det({}^t M) = \det(J) $$

SINCE $\det(J) = 1$ AND $\det({}^tM) = \det(M)$

$$(\det(M))^2 = 1$$

I.E.

$$|\det M| = 1$$

WHICH MEANS THAT VOLUMES OF PHASE SPACE ARE PRESERVED BY THE MAPPING INDUCED BY SOLUTION OF HAMILTON'S EQS. I.E. LIOUVILLE'S THEOREM

